

DRY-RUN GUIDE | interchange_diagonal.py
Source: Python\3rd_Belt\Matrix\interchange_diagonal.py

CORE BEHAVIOR

Swaps the main and anti-diagonal entry in each row of a square matrix, then prints it.

REPRESENTATIVE INPUT

2x2: '1 2' / '3 4'.

STEP TRACE

Row 0 swap -> [2,1]; row 1 swap -> [4,3].

EXPECTED OUTPUT / RETURN

Original '1 2' / '3 4'; after '2 1' / '4 3'.

ACTUAL SOURCE EXCERPT

Verbatim excerpt from Python\3rd_Belt\Matrix\interchange_diagonal.py; line numbers refer to the original file. No source was modified.

```
001 | def interchange_diagonals(matrix):
002 |     n = len(matrix)
003 |     for i in range(n):
004 |         # Swap the main diagonal element with the anti-diagonal element
005 |         matrix[i][i], matrix[i][n - 1 - i] = matrix[i][n - 1 - i], matrix[i][i]
006 |
007 | def print_matrix(matrix):
008 |     for row in matrix:
009 |         print(" ".join(map(str, row)))
010 |
011 | # Step 1: Get the dimensions of the square matrix
012 | n = int(input("Enter the size of the square matrix (N x N): "))
013 |
014 | matrix = []
015 | print(f"Enter the elements row by row (space-separated integers):")
016 |
017 | # Step 2: Dynamically take matrix input row by row
018 | for i in range(n):
019 |     row = list(map(int, input(f"Row {i + 1}: ").split()))
020 |     # Validate if the user entered exactly N elements per row
021 |     while len(row) != n:
022 |         print(f"Invalid input! Please enter exactly {n} elements.")
023 |         row = list(map(int, input(f"Row {i + 1}: ").split()))
024 |     matrix.append(row)
025 |
026 | print("\nOriginal Matrix:")
027 | print_matrix(matrix)
028 |
029 | # Step 3: Swap the diagonals
030 | interchange_diagonals(matrix)
031 |
032 | print("\nMatrix after interchanging diagonals:")
033 | print_matrix(matrix)
034 |
```

DRY-RUN NOTE

Values are representative. For function-only modules the guide uses an illustrative

direct call and explains its return; quirks/errors describe actual behavior.